Real-Time SAR Ship Detection: A Speckle-Model-Based DSP Pipeline

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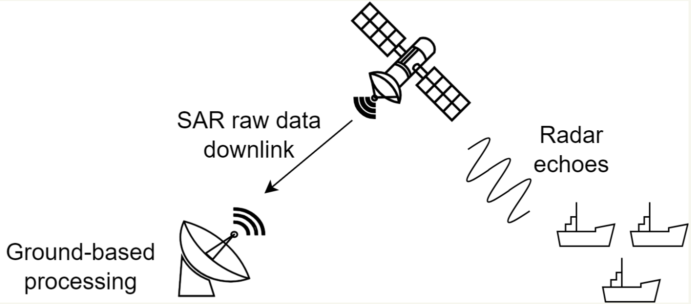
Presented by: Jen-Tse Chen
Advisor: Pei-Yun Tsai

Outline

- Introduction
- Problem and Motivation
- Highlight of Contribution
- System Definition
- Previous Methods
- Proposed Method
- Comparison
- Conclusion

Introduction

- SAR raw data are downlinked for ground-based processing, including SAR focusing, image enhancement, and ship detection.
- Detected ships provide information for maritime surveillance.

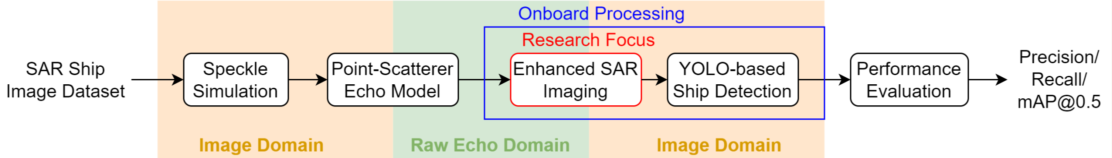


Problem and Motivation

- Ground-Based Processing
  - Large raw-data volume
  - Limited downlink bandwidth
  - Delayed target information
- Onboard Processing
  - Reduced data transmission and detection latency
  - Challenge: Limited computation, memory, and power
- Research Goal
  - Lightweight SAR processing for low-latency ship detection

Evaluation Framework

- Research Focus: Enhanced SAR imaging
- Evaluation Task: YOLO-based ship detection

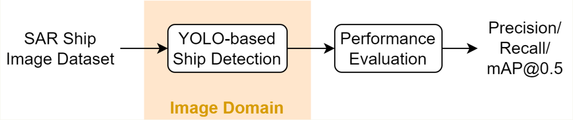


- Evaluation Metrics
  - Precision: $\frac{TP}{TP+FP}$
  - Recall: $\frac{TP}{TP+FN}$
  - mAP@0.5: Mean AP across classes at IoU threshold = 0.5

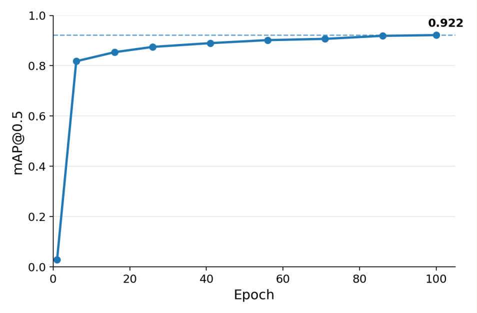
TP: True Positive, FP: False Positive, FN: False Negative

Dataset and Detection Model Setup

- HRSID
  - High-Resolution SAR Images Dataset
  - Training: 3,642 images
  - Validation: 1,962 images
  - Image size: 800 $\times$ 800 pixels
- YOLO Detection Model
  - YOLO11m fine-tuned on HRSID (100 epochs)



- Validation Performance
  - Precision: 0.910
  - Recall: 0.840
  - mAP@0.5: 0.922

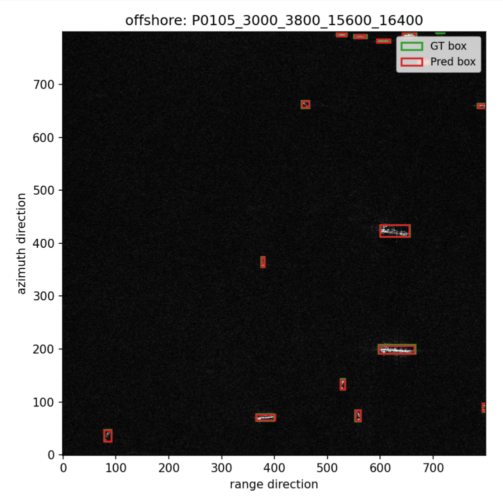


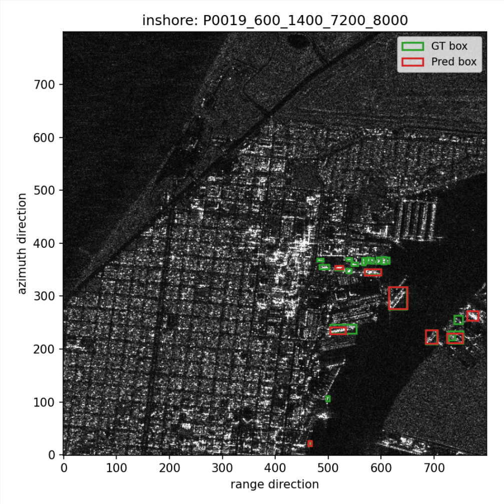
YOLO Ship Detection Results (1)

	Overall	Inshore	Offshore
Images	1,962	369	1,593
Precision	0.913	0.803	0.982
Recall	0.846	0.741	0.965
mAP@0.5	0.918	0.817	0.984

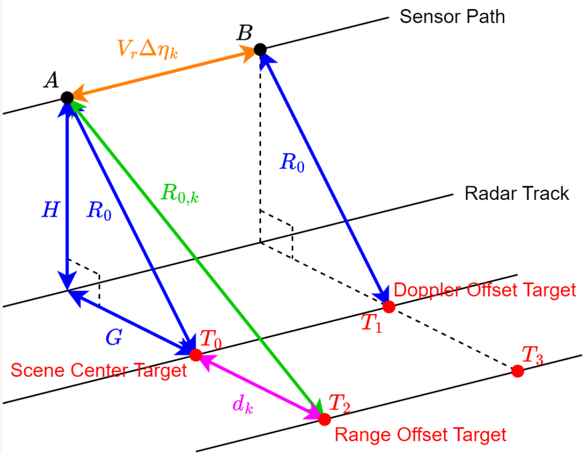
- Offshore scenes achieve better detection performance than inshore scenes.

YOLO Ship Detection Results (2)





Point-Scatterer Echo Model: Geometry



- Target position determines the time-varying slant range $R_k(\eta_m)$.
- $R_k(\eta) = \sqrt{R_{0,k}^2 + (\eta_m + \Delta\eta_k)^2 V_r^2}$
  - $R_{0,k} \approx R_0 + d_k$
  - R_0: slant range to scene center
  - d_k: range offset of scatterer k
  - $\eta_m = m\Delta\eta$: slow time
  - $\Delta\eta_k$: azimuth-time offset of scatterer k

Point-Scatterer Echo Model: Signal Synthesis

- Geometry-Dependent Delay and Phase
  - $\tau_k(\eta_m) = \frac{2R_k(\eta_m)}{c}$
  - $\phi_k(\eta_m) = -4\pi\frac{R_k(\eta_m)}{\lambda}$
- Echo Synthesis: $S = [s_{m,n}]_{M\times N}$
  - $s_{m,n} = \sum_k A_k \,\mathrm{rect}\left(\frac{\tau_n - \tau_k(\eta_m)}{T_r}\right) \cdot e^{j\pi K_r(\tau_n - \tau_k(\eta_m))^2} \cdot e^{j\phi_k(\eta_m)}$
  - $\eta_m = m\Delta\eta$: slow time, $\tau_n = n\Delta\tau$: fast time
  - A_k: scattering coefficient of scatterer k, $K_r = \frac{B_r}{T_r}$

Parameters

Compressive Sensing Algorithm

Prediction Result

Slicing

Conclusion

Reference